NO CAP

User Flow and Experience Map

Design it. Break it. Scale it. — for one learner, at zero cost.

DOCUMENT	**User Flow and Experience Map**
VERSION	**v2.1 (personal app, zero-cost)**
DATE	**16 August 2026**
SCOPE	**Core flows + optional flows by tier**
STATUS	**Working spec**

PRODUCT THESIS

A personal system-design workbench that turns passive reading into a repeatable loop of learning, visualization, reasoning, simulation, architecture practice, and interview readiness — with no paid dependencies and no accidental billing, on Cloudflare's genuine free tier, where core functionality remains usable at free-tier exhaustion.

What Changed: v2.0 to v2.1

The core flows (Daily Dose, Concept Learning, Roadmap, Review, Lab, Playground, Case Study, Interview) are **unchanged** from v2.0 — they were already right. What changes in v2.1 is that the v2.0 additions (Notes, Collections, Code Sandbox, Voice, Cost, Career, Collab) are explicitly tagged as Tier B/C/D and resequenced. First Launch is simplified: no onboarding flow (single user), no calendar connect in MVP, no target role in MVP.

Flows Preserved (Tier A)

- Daily User Flow (without Voice + Cost micro-question).
- Concept Learning Flow (without Notes + Cost section).
- Roadmap Exploration (4 modes — Career view deferred).
- Review Flow (without hands-free mode).
- Interactive Lab Flow (without cost/hr).
- Architecture Playground Flow (solo + validation only).
- Case Study Flow (8 steps — Cost + Interview Summary deferred).
- Interview Flow (text mode only — Voice deferred).
- Progress and Mastery Flow.
- Project Flow (without cost optimization milestone).

Flows Resequenced

- Notes & Annotations flow → Tier B (v1.0).
- Collections flow → Tier B (v1.0, no sharing in MVP).
- Code Walkthrough flow → Tier B (v1.0, replaces Code Sandbox).
- Voice / Audio Mode flow → Tier C (v2.0, browser TTS).
- Cost Calculator flow → Tier B (v1.0, bundled pricing).
- Career Path flow → Tier B (v1.0).
- Collaborative Design flow → Tier B (v2.0, Durable Objects).
- AI Mentor flow → Tier C (v2.0, Workers AI + quota).
- Calendar flow → Tier C (v2.0, optional).
- First Launch simplified — no onboarding, no calendar, no role.

> **EXPERIENCE PRINCIPLE (UNCHANGED)**
> NO CAP should always answer three questions for the user: What should I do now? Why is this useful? What do I do after this? The primary flow is intentionally cyclical rather than linear. Tier B/C features feed the same loop when present — they are not side-quests. Tier A MVP works perfectly without any of them.

1. Experience Strategy

NO CAP should always answer three questions for the user: What should I do now? Why is this useful? What do I do after this? The primary flow is intentionally cyclical rather than linear. v2.1 keeps the loop unchanged but prunes optional flows to later versions.

ONBOARD → BASELINE → DAILY DOSE → PRACTICE → APPLY → REVIEW → REASSESS → NEXT CONCEPT

2. First Launch

```
Open app (first time)
  |
  v
Welcome screen (1 screen, explains the loop)
  |
  v
GitHub OAuth login (single user)
  |
  v
Choose goal: interview / foundations / cloud / build / staff-promo
  |
  v
Choose level: new / familiar / strong
  |
  v
Optional baseline quiz (5 min, can skip)
  |
  v
Land on Today

  --- Deferred to later versions: ---

  Choose target role (Tier B, v1.0)
  Connect calendar (Tier C, v2.0)
```

First Launch flow (simplified for single-user app).

v2.1 simplifies First Launch dramatically. There is no onboarding flow to design (single user, GitHub OAuth). There is no roles system. There is no admin panel. The user picks a goal and a level, optionally takes a 5-minute baseline, and lands on Today. Target role selection (needed for Career Path) and calendar connection are deferred to when their features ship.

3. Onboarding Requirements

- No long profile form. Goal and level are the only required fields.
- Ask goal (interview / foundations / cloud / build / staff-promo) and current confidence (new / familiar / strong).

- Baseline is optional, takes under 5 minutes. Cover 5 concept families: Fundamentals, Scaling, Databases, Caching, Distributed Systems.
- Allow skip; the system can learn from behavior later. Skipped baseline = "unknown" mastery state, not "weak".
- Explain the learning loop in one screen (visual, not text wall).
- Land on Today, not Roadmap. Today answers "what do I do now"; Roadmap answers "where am I going" - answer the urgent question first.
- No target role prompt in MVP (Tier B, v1.0).
- No calendar connection in MVP (Tier C, v2.0).
- No voice-mode demo in MVP (Tier C, v2.0).

4. Daily User Flow [Tier A]

```
HOME
  |
  +--> Review due? ---- yes ----> REVIEW
  |                      |
  |                      v
  |                    return score
  |
  +--> Today's Dose
       |
       +--> Concept intro
       |
       +--> Visual
       |
       +--> Interactive
       |
       +--> Quiz
       |
       +--> Recall
       |
       +--> Complete
            |
            +--> streak / mastery update
            |
            +--> next recommendation

  --- When Tier B/C features ship: ---

  +--> Review due? ---- yes ----> REVIEW
  |                      |
  |                      v
  |                    hands-free flashcards (Tier C, v2.0)
  |
  +--> Today's Dose
       |
       +--> [Voice mode?] (Tier C, v2.0) --> AUDIO PLAYER
       |
       +--> ... (standard flow) ...
       |
       +--> Cost micro-question (Tier B, v1.0)
```

Daily user flow. Tier A MVP first; additions when shipped.

5. Concept Learning Flow [Tier A core]

1. User opens a concept. System shows estimated time, prerequisites, related concepts, and where it is used.

2. User enters Learn mode. Lesson blocks render progressively with Frosted Glass cards per section.

3. User can jump to Visualize, Lab, or Glossary without losing place - all open as overlays.

4. At the end, a short check verifies comprehension (3-5 questions).

5. The system updates mastery dimensions and schedules review.

6. User gets one actionable next step: continue roadmap, repeat lab, or review later.

7. **[Tier B, v1.0]** User can highlight any text or annotate any paragraph. Highlights use 4 colors mapped to learner-defined categories.

8. **[Tier B, v1.0]** A Cost Implications section shows estimated monthly cost for an architecture using this concept at reference scale.

9. **[Tier B, v1.0]** Notes written on this concept are added to the personal knowledge graph (visible in Progress > Notes Synthesis).

10. **[Tier B, v1.0]** A "View Code Walkthrough" CTA opens the Code Walkthrough Mode for implementation-heavy concepts.

6. Roadmap Exploration Flow [Tier A]

```
Roadmap
  |
  +--> Guided path
  |       |
  |       +--> next best concept (with prerequisite hints)
  |
  +--> Explore graph
  |       |
  |       +--> click concept
  |       +--> inspect prerequisites
  |       +--> jump into lesson
  |
  +--> Mastery view
  |       |
  |       +--> find weak areas (heat-map)
  |       +--> start remediation
  |
  +--> Interview view
          |
          +--> highlights interview-relevant concepts
          +--> shows frequency in real interviews

  --- Tier B (v1.0) addition: ---

  +--> Career view
          |
          +--> overlay role expectations on concept graph
          +--> gap analysis vs target role
          +--> generate 4-week gap-closing plan
```

Roadmap exploration. Career view is Tier B.

7. Review Flow [Tier A]

```
REVIEW HUB
  |
  +--> Quick review
  |      card -> answer -> confidence -> next
  |
  +--> Deep review
  |      concept -> example -> quiz -> reschedule
  |
  +--> Weak area
         concept cluster -> guided remediation

  --- Tier C (v2.0) addition: ---

  +--> Hands-free review
         |
         +--> voice prompts via browser TTS (Web Speech API)
         +--> swipe gesture: left = forgot, right = easy
         +--> hands stay free; eyes optional
         +--> ideal for commute / walking
```

Review flow. Hands-free mode is Tier C.

8. Interactive Lab Flow [Tier A]

1. User opens a lab from a concept or Practice tab.

2. The system loads deterministic initial state with default parameters. All computation is client-side.

3. User changes one or more parameters. Live metrics (latency, error rate, throughput) update in real time.

4. User is prompted to predict before revealing explanation on selected labs ("What do you think will happen to p99 if we kill Node B?").

5. User completes the objective; outputs are scored against the lab rubric.

6. Result links back to the concept and updates Application mastery dimension.

7. **[Tier B, v1.0]** Cost/hr readout uses bundled pricing data. Feeds Cost Reasoning mastery dimension (added in v1.0).

8. **[Tier B, v1.0]** Optional: "Open in Architecture Playground" creates a canvas pre-populated with the lab's architecture.

9. Architecture Playground Flow [Tier A core]

```
Choose challenge
  |
  v
Read requirements + constraints
  |
  v
Build architecture (drag, connect, configure node properties)
  |
  +--> Validate
  |      |
  |      +--> hard errors (SPOF, public DB exposure, etc.)
  |      +--> hints (learning mode only)
  |      +--> trade-off prompts
  |
  +--> Submit
         |
         v
       Deterministic rubric
         |
         v
       Reference architecture (side-by-side compare)
         |
         v
       Compare + reflect

  --- Tier B (v1.0) additions: ---

  +--> live cost annotation updates as nodes change
  +--> Export (PNG, Mermaid, Excalidraw, embed iframe, portfolio PDF)
  +--> Save to collection
  +--> Open in Cost Calculator
  +--> Version history (snapshot on every save, side-by-side compare)

  --- Tier B (v2.0) additions: ---

  +--> Invite to collab (Durable Objects + WebSockets)
  +--> Live cursors of other participants
  +--> Inline comments (anchored to nodes)
```

Architecture Playground flow. Tier A MVP first.

10. Architecture Reflection Flow

After every architecture challenge, prompt the learner with three questions: "What did you optimize for?", "What did you sacrifice?", and "What fails first?" Answers are stored as short notes attached to the attempt and feed the Explanation mastery dimension. **[Tier B, v1.0]** adds a fourth question: "What does it cost per month at the target scale, and what is the dominant cost driver?" - which feeds the Cost Reasoning dimension. The product rewards reasoning quality, not only node selection.

11. Case Study Flow [Tier A core]

```
Case Study: Design WhatsApp
  |
  01 Requirements
  02 Capacity estimation
  03 APIs
  04 Data model
  05 HLD
  06 Realtime delivery
  07 Reliability
  08 Failure modes
  09 Trade-offs

  --- Tier B (v1.0) additions: ---

  10 Cost estimation (uses bundled pricing dataset)
  11 Interview summary (90-second narrative, scored for coverage + clarity)
```

Case study stepper. Cost + Interview Summary steps are Tier B.

Tier A MVP ships 9 steps. Tier B (v1.0) adds: Cost Estimation (learner estimates monthly run cost at the estimated scale, then compares against a reference cost using bundled pricing data) and Interview Summary (learner writes a 90-second narrative they could deliver at the end of an interview; system scores for coverage and clarity).

12. Case Study Two-Lane Mode

Lane	Behavior
Guided	Step-by-step hints, explanations, reference snippets. Cost step (when shipped) shows the reference cost breakdown.
Challenge	User designs first, reference architecture and cost are hidden until submit. Submit triggers full rubric scoring.

13. Interview Flow [Tier A text, Tier C voice]

```
INTERVIEW HOME
  |
  +--> Choose prompt
  |      URL shortener / Instagram / custom
  |
  +--> Choose duration
  |      30 / 45 / 60 minutes
  |
  +--> Interview (Tier A: text mode)
         |
         +--> requirements
         +--> estimation
         +--> HLD
         +--> deep dive
         +--> interviewer follow-up (text)
         +--> final trade-offs
         |
         v
       score + rubric
         |
         +--> weaknesses
         +--> recommended concepts
         +--> retry interview

  --- Tier C (v2.0) addition: Voice Interviewer mode ---

  +--> Choose mode: text | voice
  |      |
  |      +--> voice: AI interviewer asks aloud (browser TTS)
  |                    learner speaks answers (browser STT)
  |                    final rubric includes communication clarity score
```

Interview flow. Voice mode is Tier C.

14. AI Mentor Flow [Tier C, v2.0]

```
User action
  |
  +--> Explain this
  +--> Quiz me
  +--> Give hint
  +--> Critique my architecture
  +--> Act as interviewer
         |
         v
      Check Workers AI quota
         |
         +-- quota OK --> AI mentor session
         |                  |
         |                  v
         |               curated concept context
         |                  |
         |                  v
         |               explanation / feedback
         |                  |
         |                  v
         |               user decides whether
         |               to turn insight into review
         |
         +-- quota exhausted --> Authored fallback
                                   |
                                   v
                               "AI quota reached for today.
                                Authored explanation below."
```

AI Mentor flow with quota guard and fallback.

AI Mentor is Tier C (v2.0). It uses Cloudflare Workers AI with the free 10k-neurons/day allocation. Every request is wrapped in a quota guard: when daily neurons are exhausted, the mentor UI shows "AI quota reached for today - authored explanation below" and falls back to deterministic content. The learner sees current quota usage in the mentor panel. The app works perfectly without AI — AI is an enhancement, never a dependency.

15. AI Safety / Product Boundary

· AI output never silently becomes official course content.

· AI suggestions cannot override deterministic assessment without user action.

· The UI labels AI-generated guidance clearly with a small "AI" chip in amber.

· When Workers AI quota is exhausted, the UI shows "AI quota reached for today" and offers authored fallbacks. No paywall, no upgrade prompt.

· Users can delete AI chat history at any time. Deletion is irreversible.

· AI cannot read private notes unless the user explicitly shares them in a mentor session.

- Voice interviews (Tier C): browser-native STT transcribes locally where possible; transcripts stored in D1 are deletable.
- AI never generates cost estimates directly - it can only explain the output of the deterministic cost engine (which uses bundled pricing data).

16. Project Flow [Tier B]

```
Project catalog
  |
  +--> Choose build (e.g., "Build a rate limiter", "Build a tiny queue")
  |
  +--> Requirements
  |
  +--> Milestones
  |      1. baseline (single-machine)
  |      2. scale (10x traffic)
  |      3. reliability (kill a node, survive)
  |      4. observability (instrument metrics + tracing)
  |
  +--> Submit evidence (architecture + metrics)
  |
  +--> Architecture review (self-review against rubric)
  |
  +--> Reflect
  |
  +--> Mark project complete

  --- Tier B (v1.0) addition: ---

  5. cost optimization (estimate cost, propose 3 optimizations)
```

Project flow. Cost optimization milestone is Tier B.

17. Progress and Mastery Flow

The user should be able to move from any weak area directly into remediation. Example: Progress → Reliability weak → see top weak concepts → choose Circuit Breaker → review → simulation → architecture challenge. **[Tier B, v1.0]** adds two new remediation paths: Cost weak → Cost Calculator walkthrough; Career gap → 4-week gap-closing plan. **[Tier A, v2.1]** adds the Quota Dashboard widget so the learner can see daily Cloudflare free-tier usage.

18. Notification / Return Flow

Trigger	Experience	Tier
Review due	In-app badge on Review nav item. PWA push notification (opt-in).	A
Streak at risk	In-app banner on Home. No push (avoids annoyance).	A
Milestone reached	Brief celebration modal. Never auto-share.	A
Quota warning	When daily Worker/D1/AI usage hits 80%, show subtle banner: "Heads up: 80% of today's quota used."	A
Interview date approaching	Shift recommendations toward interview-relevant skills. Countdown appears on Home.	A (date set manually)
Project waiting	Resume exact milestone state. "You left the reliability milestone half-done."	B
Cost data refreshed	Quiet badge on Cost Calculator icon. (Quarterly dataset refresh via repo PR.)	B
Weekly digest	Optional in-app summary: concepts mastered, time spent, weak areas, recommended next.	B
Calendar-based session reminder	Fires 5 min before scheduled learning session. Deep-links to Today.	C
Collab invite	In-app modal with "Join canvas" button.	B (v2.0)

19. Notes & Annotations Flow [Tier B, v1.0]

```
In any lesson or concept page
  |
  +--> Select text
  |      |
  |      +--> Highlight (4 colors = 4 learner-defined categories)
  |      +--> Add margin annotation (short scribble, anchored to selection)
  |      +--> Convert to flashcard (auto-creates a review item)
  |
  +--> Tap "Notes" tab on concept page
  |      |
  |      +--> View all notes for this concept
  |      +--> Add longer free-form concept note
  |      +--> Link to other concepts (bidirectional)
  |
  +--> Notes appear in:
         |
         +--> Progress > Notes Synthesis (knowledge graph)
         +--> Search (Cmd+K includes notes)
         +--> Review (converted flashcards)
         +--> AI Mentor (optionally shareable, Tier C)
```

Notes & Annotations flow (Tier B).

Notes are private by definition (single user). The knowledge graph visualization in Progress > Notes Synthesis renders concepts as nodes and personal notes as edges — showing where the learner has built the densest understanding and where they have gaps. This is one of the most powerful feedback loops in v1.0: it makes the learner's own synthesis visible to them.

20. Collections & Study Lists Flow [Tier B, v1.0]

```
Collections hub
  |
  +--> Create new collection
  |      |
  |      +--> Name + optional goal (interview / refresher / role-based / custom)
  |      +--> Add concepts (search or pick from roadmap)
  |      +--> Reorder via drag
  |
  +--> Open existing collection
  |      |
  |      +--> Study mode (sequential concept walk-through)
  |      +--> Quiz mode (auto-generated quiz from collection concepts)
  |      +--> Flashcard mode (auto-generated from notes + highlights)
  |
  +--> System starter collections
         |
         +--> "Google L5 interview pack" (curated, ~30 concepts)
         +--> "AWS SAA prep" (curated, ~25 concepts)
         +--> "Foundations refresher" (auto-built from your weak areas)
         +--> "Staff eng breadth pack" (curated, ~40 concepts)
```

```
--- Tier B (v2.0) addition: ---

+--> Share collection (read-only link, via R2 presigned URL)
```

Collections & Study Lists flow (Tier B).

Collections turn NO CAP from a linear curriculum into a personal study toolkit. A learner prepping for a Google L5 interview can build a custom pack of the 30 concepts most relevant to that interview, study them in their preferred order, auto-generate quizzes and flashcards. The system starter collections lower the activation energy — one tap to start a focused prep track. **Sharing is deferred to v2.0** (single-user app in MVP).

21. Code Walkthrough Flow [Tier B, v1.0] (replaces Code Sandbox)

```
Concept page > "View Code Walkthrough"
  |
  v
Code Walkthrough Mode opens (single pane, no execution)
  |
  +--> Read code (with syntax highlighting + inline comments)
  |
  +--> Explain line (click any line, see authored explanation)
  |
  +--> Modify code (edit in editor, no execution)
  |      |
  |      v
  |    "Predict output" prompt
  |      |
  |      v
  |    Learner reasons about output
  |      |
  |      v
  |    Reveal authored expected output + explanation
  |
  +--> Identify bottleneck (find the slow line)
  |
  +--> Map to architecture (open Architecture Playground pre-populated)
```

Code Walkthrough flow. No live execution - read, modify, predict only.

Code Walkthrough Mode replaces the v2.0 Code Sandbox. The educational value is in **reading, modifying, and predicting** — not in executing. The learner can edit the code, but "running" it shows an authored expected output, not a live execution. This delivers 90% of the educational value at 0% of the infrastructure cost. Live execution (the original Code Sandbox) is Tier D — deferred until constraints change.

22. Voice / Audio Mode Flow [Tier C, v2.0]

```
Any lesson or Daily Dose > Voice toggle
  |
  v
Audio player with chapter markers (browser-native TTS)
  |
  +--> Listen (Web Speech API: SpeechSynthesis)
  |      |
  |      v
  |    Voice quality depends on browser/OS:
  |      Chrome on macOS: excellent
  |      Safari on iOS: good
  |      Some Linux browsers: limited
  |      |
  |      v
  |    Always offer "Show text alongside" fallback
  |
  +--> Scrub via chapter list
```

```
|
+--> Adjust speed (0.8x to 2.0x)
|
+--> Hands-free flashcards
|       |
|       v
|     Voice prompts question via TTS
|     Learner speaks answer (or swipes)
|     Swipe gestures: left=forgot, right=easy
|
+--> Voice interview practice (Tier C)
        |
        v
    AI interviewer asks questions aloud (browser TTS)
    Learner speaks answers (browser STT: SpeechRecognition)
    Transcript stored in D1, deletable
    Final rubric includes communication clarity score

--- Tier D (future) addition: ---

+--> External TTS (ElevenLabs-quality) via paid provider
     (only if user explicitly opts in and provides API key)
```

Voice / Audio Mode flow using browser-native APIs.

Voice Mode uses the **Web Speech API** — browser-native, zero server cost, zero quota. Voice quality varies by browser/OS. The UI always shows the current voice and offers a "show text alongside" fallback. **No paid TTS provider is used in v2.1.** External TTS (ElevenLabs-quality) is Tier D — deferred until constraints change, and only if the user explicitly opts in with their own API key.

23. Cost Calculator Flow [Tier B, v1.0]

```
From any architecture (saved or in-progress) or case study
  |
  v
Cost Calculator opens
  |
  +--> Pricing dataset: AWS (bundled, "as of 2026-08-01")
  +--> Adjust node properties (instance type, reserved vs on-demand, etc.)
  |
  +--> View breakdown table (service x qty x $/mo x % of total)
  +--> View dominant cost driver (highlighted in amber)
  +--> View cost per user / per request / per transaction
  |
  +--> Save cost estimate to architecture
  +--> Add to case study (as the Cost step answer)
  +--> Export cost report (PDF or CSV)

  --- Note: NO live cloud pricing APIs in v2.1 ---

  Pricing dataset is:
    - bundled in the git repo as versioned JSON
    - refreshed quarterly via PR
    - ~20 common AWS services initially
    - GCP and Azure added later (Tier B, v1.0+)
    - "as of" date shown on every estimate
    - works offline, costs zero Worker requests
```

Cost Calculator flow with bundled pricing dataset.

24. Career Path Flow [Tier B, v1.0]

```
Progress > Career Path
  |
  +--> Select target role (L3 / L4 / L5 / Staff / Principal)
  |
  +--> View gap analysis
  |      |
  |      +--> Per-family mastery bar vs role expectation
  |      +--> Top 3 gaps prioritized
  |
  +--> Generate 4-week gap-closing plan
  |      |
  |      v
  |    Focused roadmap with weekly milestones
  |    Each week targets one of the top 3 gaps
  |
  +--> Track progress (plan completion %)
  |
  +--> Switch target role anytime (regenerates plan)
```

Career Path flow (Tier B).

25. Collaborative Design Flow [Tier B, v2.0]

```
Architecture Playground > Invite
  |
  +--> Generate joinable link (32-byte random slug)
  |
  +--> Invitee opens link
  |      |
  |      v
  |    Durable Object creates participant slot
  |    CRDT doc loaded from R2 snapshot (or fresh)
  |
  +--> Concurrent editing (Yjs CRDT, conflict-free)
  |      |
  |      +--> Live cursor presence (colored, named)
  |      +--> Inline comments (anchored to nodes)
  |
  +--> Owner can lock nodes (for instructor-led sessions)
  |
  +--> Session ends: snapshot saved to R2, all participants get a copy

  Infrastructure:
    - Durable Objects: presence + CRDT sync (free tier: 100k req/day)
    - R2: snapshot storage (free tier: 10GB, free egress)
    - No Redis, no Yjs-server, no S3
    - Max 4 concurrent editors (personal use)
    - Auto-end after 2h inactivity
```

Collaborative Design flow (Tier B, v2.0). Durable Objects-based.

Collaborative Design uses Durable Objects for presence and CRDT sync, R2 for snapshots. No Redis, no Yjs-server, no S3 — all Cloudflare free tier. Max 4 concurrent editors (personal use, not a Miro replacement).

26. Failure / Edge Flows

· Network unavailable -> PWA service worker serves cached core shell + current lesson + notes. Static content bundles work fully offline.

· AI quota exhausted -> show "AI quota reached for today" + authored fallback. No paywall, no upgrade prompt.

· Browser TTS unavailable -> show "Browser TTS not available. Text version below." Voice mode simply not offered.

· Browser STT unavailable -> voice interviewer mode not offered. Text interview mode works.

· Bundled pricing data stale -> show "as of [date]" badge. If stale > 90 days, show "pricing data may be outdated - consider refreshing the dataset" warning.

· Quiz submission conflict -> preserve local answer and retry with exponential backoff. Never lose learner input.

· Session expiry -> re-auth via GitHub OAuth without losing in-progress local state. Unsaved work is preserved in localStorage and re-synced on re-auth.

· Corrupt saved architecture -> load last valid snapshot from D1. Offer "Restore from version history" picker (Tier B).

· D1 quota near limit -> quota dashboard warns at 80%. Defer non-essential writes (e.g., analytics events) until next day.

· Worker quota near limit -> quota dashboard warns at 80%. Defer non-essential requests (e.g., AI mentor) until next day.

· Collab sync conflict -> CRDT handles merge automatically; if a node was edited by two users simultaneously, both versions appear in version history.

27. Mobile-Specific Flow

v2.1 prioritizes web/PWA. Mobile is "read and review" only initially: Daily Dose, Review, Flashcards, Progress, Audio (when shipped). Deep architecture work, case studies, and the playground stay on desktop. PWA is installable on mobile home screen for app-like experience without app store overhead. Native mobile app (React Native) is Tier D — deferred until the web/PWA is solid and constraints change.

Lock-screen widgets (iOS + Android) showing daily streak and review-due count are nice-to-have but require native shell — Tier D. PWA can use the Badging API (where supported) for a lightweight review-due indicator on the home screen icon.

28. User Journey: Beginner to Interview Ready

Week	Focus	Key Activities	Version
Week 1	Fundamentals + Daily Dose	First launch + baseline, Daily Dose habit, first 5 concepts. Tier A only.	0.1
Week 2-3	Scaling + DB + Caching	First interactive lab (client-side), first architecture playground attempt.	0.5
Week 4	Networking + Async + Distributed	Focus mode habit, complete first architecture challenge.	0.5
Week 5	Reliability + Observability + Security	First case study (partial, 9 steps), first project milestone.	0.5
Week 6	Cloud + Real-time + AI systems	Cloud-specific concepts, complete first case study, first text interview.	0.5
Week 7+	Case studies + Projects + Interviews	First mock interview (text mode), build first collection (Tier B).	1.0
Week 10+	Notes synthesis + Cost + Career	Cost reasoning deepens, career path planning, notes knowledge graph.	1.0
Week 14+	Voice + AI + Collab (optional)	Voice mode for commute learning, AI mentor sessions (quota-guarded), collaborative design.	2.0
Indefinite	Mastery loops continue	Multi-cloud cost comparisons (when pricing data expands), portfolio export for job search.	1.0+

29. UX Guardrails

- Never leave the user at a dead-end after completing a lesson. Always offer one next action.
- Every completion screen offers one next action only as the primary CTA. Secondary actions are smaller and below.
- Do not make the user navigate back through three screens to revisit a related concept. Use the related-concepts chips at the bottom of every concept page.
- Persist unfinished work aggressively. Quizzes in progress, architecture sketches, notes - all auto-save within 2 seconds of change.
- Keep challenge feedback specific: what happened, why, and what to try next. Never "Try again" alone.
- Make difficulty visible before the user starts an assessment. Show estimated time and tier (Warm-up / Solid / Hard / Interview / Staff).

- Never auto-share anything. (Moot in single-user MVP, but the rule holds for v2.0 collab.)
- Notifications are gentle by default. The only "urgent" notification is quota-warning at 80%.
- Voice mode is always optional. Never auto-play audio. Always show a text equivalent.
- Cost data is always dated. Never show a cost estimate without an "as of [date]" badge.
- Quota guards never produce bills. When a quota is hit, the feature becomes unavailable with a friendly explanation — never a paywall.

29. Critical End-to-End Acceptance Flow

This flow describes a full Tier A + Tier B session (the v1.0 product). Steps marked [C] require Tier C features; substitute the deterministic fallback when those are unavailable.

1. New user signs in via GitHub OAuth, selects interview-prep goal, level "familiar".

2. Baseline establishes weak foundations in caching and distributed systems.

3. Home recommends CAP fundamentals with a "high interview frequency" badge.

4. User completes Daily Dose (12 min standard). [C] Tries Voice mode for the recall step.

5. Review is scheduled for the correct future date.

6. User practices a CAP simulation (client-side). [B] Cost/hr readout updates as they tune parameters. They save the cost estimate to the architecture.

7. User attempts a replication challenge in the Architecture Playground. [B] They highlight a paragraph in the lesson and write a margin note.

8. Progress dashboard updates the mastery matrix. [B] Cost Reasoning dimension shows first data point. Quota dashboard shows today's Worker/D1 usage.

9. A case study (Design WhatsApp) uses those concepts. [B] The Cost step compares learner estimate ($3.8k/mo) to reference ($4.2k/mo) - 90% accuracy.

10. Interview mode (text) surfaces the same weak concepts in follow-up prompts. [B] The Interview Summary step records a 90-second narrative.

11. [C] AI mentor explains a gap but does not alter official mastery directly. Quota guard shows 60% of today's neurons used. User adds the AI explanation to their notes.

12. User receives a clear recommendation for the next concept. [B] The 4-week gap-closing plan (Career Path) updates to reflect today's progress.

13. [B] User exports their portfolio (architectures + mastery matrix + project evidence) as a PDF to attach to job applications.

14. Quota dashboard confirms: 12k Worker requests used today (of 100k), 340k D1 rows read (of 5M), 1.2k AI neurons used (of 10k). All well within free tier. Cost: Rs 0.

30. Final Experience Principle

> **THE LOOP (REFINED FOR V2.1)**
> Every screen should feed one of seven actions: learn, understand, practice, apply, recall, reason about cost, or explain aloud. If a feature cannot improve one of those actions, it is probably not core. Tier A MVP ships with five (learn, understand, practice, apply, recall). Tier B adds cost reasoning. Tier C adds voice explanation. The product is the loop — everything else is in service of it.